ISyE 6501
Introduction to Analytics Modeling
Course Project

This project should be done individually.

The web sites https://www.sas.com/en_us/customers.html, https://www.ibm.com/case-studies/search?search, and https://www.informs.org/Impact/O.R.-Analytics-Success-Stories (among others) contain brief overviews of some major Analytics success stories. In this course project, your job is to think carefully about what analytics models and data might have been required.

1. Browse the short overviews of the projects. Read a bunch of them – they're really interesting. But don't try to read them all unless you have a lot of spare time; there are lots!
2. Pick a project for which you think at least three different Analytics models might have been combined to create the solution.
3. Think carefully and critically about what models might be used to create the solution, how they would be combined, what specific data might be needed to use the models, how it might be collected, and how often it might need to be refreshed and the models re-run. DO NOT find a description online (or elsewhere) of what the company or organization actually did. I want this project to be about your ideas, not about reading what someone else did.
4. Write a short report describing your answers to (3).

The Indiana Department of Corrections and Reduction of Violent Attacks

Please note: I will be making two assumptions in this solution. The first assumption is that the data surrounding the attacks over the last 18 months is documented, detailed and available for implementation. The second assumption is that the cost of the project and the necessary work to maintain the data will be totally covered by the amount of money the IDOC will save from reducing violent attacks.[1]

The Indiana Department of Corrections (IDOC) is a statewide prison system comprised of 18 facilities and 27,000 inmates staffed by 5,500 employees. Over an 18 month period, there was an increasing trend in inmate attacks on staff members. Across all of the IDOC 18 adult facilities, they were average 320 attacks per month, a little over 10 per day. This was an extremely serious, disturbing, and expensive problem for the IDOC. Prior to addressing the issue, the IDOC utilized the Indiana Risk Assessment System (IRAS) for risk assessment on inmates. The problem with IRAS was that it was designed to predict recidivism, or the likelihood that an inmate would reoffend and be arrested again for a similar offense. This tool was not effective for predicting the likelihood of an inmate acting out in

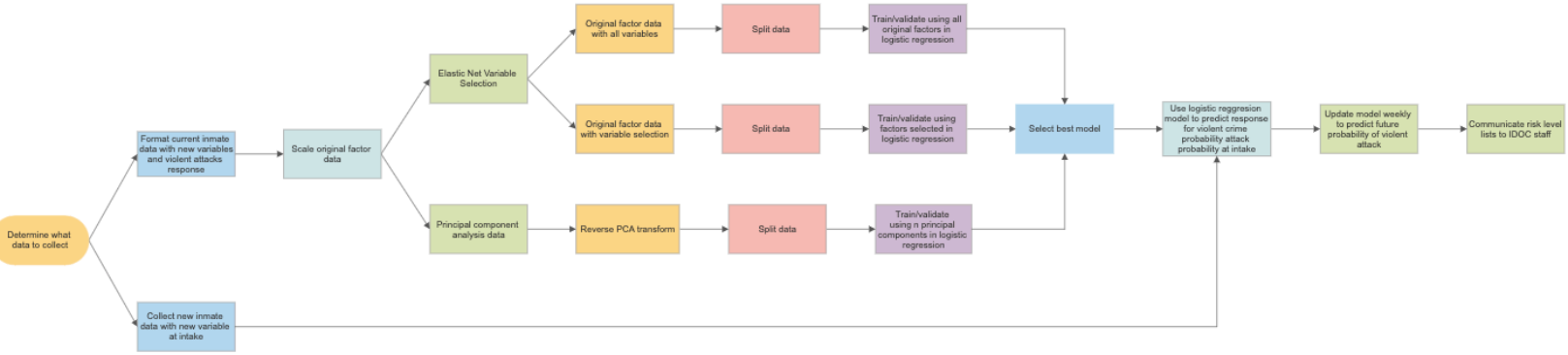
[1] "Not only was this surge in violence disturbing, it's expensive for the state in terms of workers' compensation and non-productive hours." - Sarah Schelle, Executive Director of Legislation and Data Science of the IDOC https://www.sas.com/en_us/customers/idoc.html

violence. IRAS also only used information on predictive factors at the intake portion of prison sentence and failed to model how the risk may change over the course of an inmate's sentence.

With that in mind, the IDOC set out to create a new and risk assessment tool. Per the IDOC, an effective risk assessment tool would do three things:

1. Incorporate all relevant data sources
2. More accurately predict violence
3. Clearly communicate the findings to facility staff

In the following report, I will propose one solution that utilizes a **logistic regression model**, a **survival model (Cox proportional hazards model)**, and **Louvain algorithm of community detection** that the IDOC could incorporate to effectively reduce the rate of violent attacks. The work flow of this proposed solution is pictured below.



The first step in creating the new risk assessment tool, incorporating all relevant data sources, will focus on the data collection/data preparation.

Data Collection for IDOC Risk Assessment Tool

"A model is only as good as the data it is trained on." The data collection and preparation will arguably the most important part of the risk assessment tool re-design for the IDOC. Knowing what predictors to focus on and which predictors are less significant (or possibly too costly) could make the difference between a model that is effective at assessing violent risk and an ineffective, costly and potentially dangerous waste of time. Since the current risk assessment tool uses data captured at intake and is only successful at predicting recidivism, I will be suggesting a radical change in **what** data the IDOC collects and **how** the IDOC collects, updates and stores this data.

• What data to collect:

In choosing what data to collect and what variables may be predictive of violent attacks, a great place to start would be with the experts in the field - the corrections officers and the health services division (this includes inmate services such as addiction recovery, mental health, physical health, staff

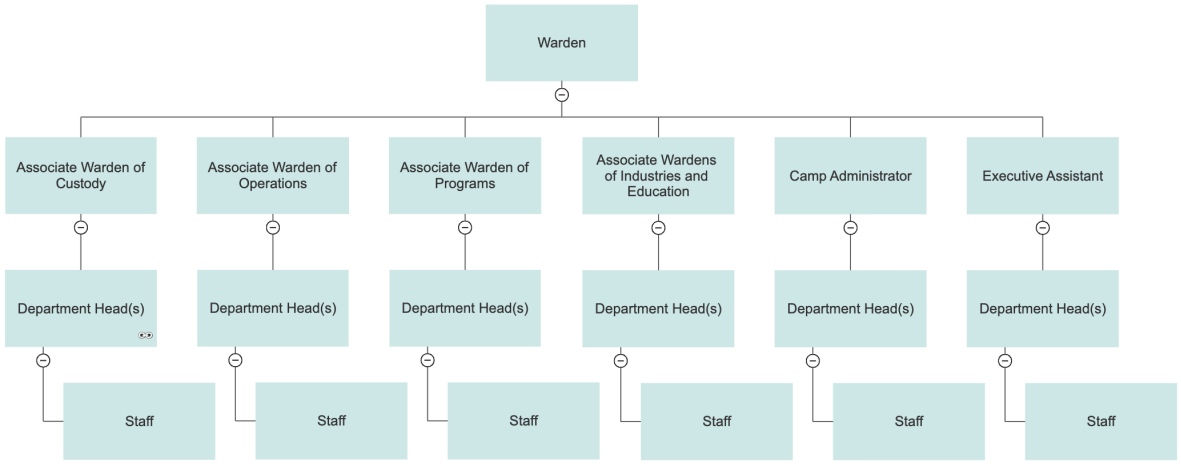
health and wellness, etc.)[2] The corrections officers and health services staffs at each facility interact daily with each of the 27,000 inmates and many of them have many years of experience in these interactions. Their insights would be invaluable in this process. Some of these variables may include, but not limited to, the following: history of confrontations with staff/other inmates, sentence for violent crime(s), known gang affiliation, anger management counseling, security level of prison (maximum, medium, minimum etc.), verbal confrontations, contraband violations, sentence length, age, among **many** others. I would also recommend reviewing the variables with the legal department in order to avoid including any possible discriminatory practices in the data collection.

• How to collect/update the data:

The IDOC will utilize a two step process for how they collect/update data - at intake and via weekly inmate reports. The intake data will serve as a sort of "baseline" for incoming inmates. It will reflect the changes in data collection noted above and will be used inputs for an initial probability that inmate will commit a violent attack. The second and most important (and more resource intensive) part of the new risk assessment tool will be consistent monitoring and updating of data and the resulting changes in violence probabilities, done using weekly inmate reports. A lot can change for an inmate on a yearly, monthly, or even weekly basis. Changes in data (and violence probabilities) need to reflect the changes in circumstances for life as an inmate. Due to the frequency of attacks (nearly 10 per day/70 per week), it would be appropriate to file weekly reports for inmates who experience **material** changes in any of the predictive factors. An example of this may be an inmate enrolled into anger management or an inmate transferred from a medium security to a maximum security facility. This level of detail is necessary for maintaining more accurate predictions and keeping staff and inmates safe.

• How to store the data:

Finally, the IDOC will be storing this data from all 18 facilities in a central, easily accessible, intuitive dashboard so that communication between facilities and even within facilities is seamless. All staff will have access to the dashboard and risk assessment tool outputs. Additionally, each facility will require weekly, chain of command communication about the current risk assessment. The information flow will be top down communication, from warden to each department staff, about the weekly risk. A generic structure of prison hierarchy and how information might flow is pictured below. [3]



[2] This information about the different departments in IDOC Health Services was obtained from https://www.in.gov/idoc/medical/

[3] This chart was created using information about prison structure obtained from https://prisonprofessors.com/chain-of-command-prison/#:~:text=Under%20the%20Unit%20Manager%2C%20for,there%20will%20be%20many%20foremen

The information must also be easy to interpret for all of the staff. Inmates will be put into color coded lists of low (green), medium (yellow) or high (red) risk inmates. The lists will be determined by probability thresholds ($p >= 0.33$ low, $0.33 < p < = 0.67$ medium, $p > 0.67$ high). These thresholds can be adjusted over time based on the performance of the risk assessment tool. Below is one example of an easily interpretable table of risk lists that would be accessible to staff, with each inmate assigned a risk score, sort from highest to lowest, and each inmate number being a link to the inmate file and notable risk factors.

Prison X Risk Assesment 04/17/2023-04/23/2023

High Risk		Medium Risk		Low Risk	
Inmate #	Risk Score (Out of 100)	Inmate #	Risk Score (Out of 100)	Inmate #	Risk Score (Out of 100)
91798	75	96063	65	52404	30
35072	73	98938	65	8726	29
28416	72	52000	65	75344	28
82399	72	5505	65	25984	27
2801	70	67651	55	24555	26
12007	70	62566	54	71486	25
52630	70	68009	45	96279	24
50482	68	63779	44	76902	23
66319	68	63503	43	29277	22
		35563	42	27461	21
		77613	41	35774	20
				48850	19
				71212	18
				64137	17
				4222	16
				94074	15
				46113	14
				52594	13
				81213	12
				42650	11
				33179	10
				32431	10
				48528	10
				74772	10
				48380	10
				43926	10
				44785	10
				36603	10
				5892	10
				89505	10
				39136	10
				86246	10
				16095	10
				16349	10

Data Preparation for IDOC Risk Assessment Tool

- Formatting current inmate data:

The data collection phase addressed the collection of new inmates and future updating of new inmates as well as current inmates for the new risk assessment tool, but there remains another issue. What about the initial collecting of current inmate data for the new risk assessment tool? Per one of the

key assumptions, the IDOC has the relevant data for the utilized potential variables. After formatting the existing data with the appropriate predictors for the new risk assessment tool, a response variable will be added to the data as well. Each existing inmate involved in a violent attack over the last 18 months will receive a 1 in the response variable and each existing inmate who was not involved in a violent attack in the last 18 months will receive a 0. This will be important in training, validating and testing of the models.

• Scaling the data:

Another key step in the data preparation phase for the risk assessment tool will be to scale the data. Depending on what predictors are selected for the new risk assessment tool, the data could have very wide ranges in magnitudes that could disproportionately affect the violent attack probabilities. For example, a predictor like age would have a 10^1 order magnitude while sentence length in days could have in the 10^4 order magnitude. Scaling the data would level the affect of these types of predictors on the outcome of violent attack probabilities. Scaling will only take place on the predictors, not the response variable.

• Creating PCA version of data:

Finally, because the data could potentially be highly dimensional, I recommend creating a version of the data that utilizes principal component analysis (PCA). PCA is a great technique for data with lots and lots of variables. On its surface, it attempts to reduce correlation in the data's many variables and ranks the components by importance (highest variance to lowest variance). Cumulative proportion of variance (rule of thumb is to use the number of components that explain >80% of the total variance), the eigenvalue test (keeping the number of components who's eigenvalues are > 1), and a scree plot (visual representation of variance explained by each principal component) can all be used to determine n amount of principal components to be included in the model. Finally, reversing the PCA transformation to the original factor space will provide coefficients for the model with a potential reduction in correlation and randomness. Because the new form of data could be highly dimensional, creating PCA data could reduce some of the variables eliminate correlation where it exists. **The models will be trained, validated, and tested using the both original factor data and the PCA data to see which model performs best.** PCA will also only take place on the predictors, not the response variable.

Model #1: Logistic Regression

GIVEN {both original factor data and PCA data on current inmates in the IDOC}
USE {logistic regression model}
TO {train/validate/test to find the best combination of model and data set. The best model/data set will be used on future inmates at intake to generate an immediate probability of violent attacks}

• GIVEN {both original data and PCA data on current inmates in the IDOC}

This is the original factor data and the PCA data as well as the response variable that was collected on the current 27,000 inmates.

• USE {logistic regression model}

I am recommending a logistic regression model. Logistic regression models estimate probability using a response variable of either 0 or 1. In this case, the model will be trained, validated and tested using the above data to find the best fit model. Logistic regression utilizes a pseudo r-squared to estimate model fit. The model can also be used to see how well (or not) current threshold of high risk (p > 0.67) predicts the response of the current inmates who were or were not involved in violent attacks.

- *TO {train/validate/test to find the best combination of model and data set. The best model/data set will be used on future inmates at intake to generate an immediate probability of violent attacks}*

Next, the original factor data will be put through variable selection. I will recommend utilizing elastic net for variable selection. Elastic net combines lasso (which does variable selection by removing statistically insignificant factors) and ridge regression (which provides predictive benefits by reducing coefficients). This should help shrink the number of variables as well as the impact of some other variables in the risk assessment tool.

This step is important for a few reasons. The simplest model is often best to avoid overfitting, decreasing risk of including insignificant variables and it is also easier to interpret/implement. If it is discovered that only 3 or 4 variables are needed to more accurately predict violent attacks, data collection becomes easier at intake, updating data/risk lists becomes less time intensive, the higher powers in the IDOC are more likely to implement the model because it is easier to explain and understand, and division of resources for the IDOC (where to put corrections officers, how to de-escalate, mental health treatment, etc) becomes more clear.

Armed with both the original data and the PCA version of the data and the response variable of attacks over the last 18 months, it is necessary to split the data into two groups - training/validating data and testing data. The groups will be split (pseudo) randomly to limit bias/randomness in the model. 80 percent of the data points will be used for training/validating and the remaining 20 percent will be set aside for testing the fit of the models. Since each inmate comprises a data point, about 21600 inmates will be in the training/validating split and the remaining 5400 inmates will be used as the testing set. This will be done for both original data and PCA data. This process may yield a table of results similar to the one pictured below.

Fit of Each Model		
Model Used	**Psuedo R-squared validated**	**Psuedo R-squared tested**
Scaled, All variables	0.51	0.46
Scaled, variable selection	0.5	0.46
Scaled, PCA	0.47	0.44

In this scenario, we would select the model that used scaled data and variable selection. This model tested as well as the model with all variables, so the simpler, easier to implement/understand model will be utilized to produce probability of violent attack as baselines for new inmates at intake.

Model #2: Survival Model (Cox Proportional Hazards Model)

GIVEN {the weekly updated data forms on current inmates who experience material change}
USE {Cox proportional hazards model with factors used in best fit model above}
TO {find probabilities of violent attacks for each day in the coming week (t+1, t+2, …t+7) and update risks for violent attacks in risk lists using highest probability in each of the next 7 days}

- GIVEN {the weekly updated data forms on current inmates who experience material change}

The data from the current inmates that reflects any material changes noted in weekly inmate reports.

• USE {Cox proportional hazards model}

Cox proportional hazards model is a survival model that predicts the probability of an event happening at time, t, using specific values of predictors. Utilizing the updated data forms and the predictors used in variable selection from the tested models, a probability for violent attacks can be estimated for each day of the week, t+1 for tomorrow all the way up to t+7 for 7 days from now, before the next round of update reports.

Cox proportional hazards models operate with a data issue. Because the event has yet to occur, the data is, at its core, censored. There are ways to work around the censored data that are a bit complex, but for the purposes of this exercise, this model's output will suffice.

• TO {find probabilities of violent attacks for each day in the coming week (t+1, t+2, …t+7) and update risks for violent attacks in risk lists using highest probability in each of the next 7 days}

A cox proportional hazards model may produce probabilities of violent attacks for the next 7 that look something like the values below.

Probability of Violence for Inmate #23478							
T+1	T+2	T+3	T+4	T+5	T+6	T+7	Max{t+1:t+7}
0.63	0.68	0.62	0.6	0.59	0.58	0.57	0.68

In this instance, Inmate #23478 had a maximum probability of violence over the next 7 days of 0.68. Taking the maximum will allow the IDOC to operate in an abundance of caution. This inmate will be placed on the high risk of violent attacks list as a result.

Model #3: Louvain Algorithm of Community Detection

GIVEN {Information about prison infrastructure and chain of command}
USE {Louvain algorithm}
TO {maximize modularity and to detect large communities and facilitate weekly information about risk assessment and risk lists}

• GIVEN {Information about prison infrastructure and chain of command}

Information about the hierarchy of the prison, top down chain of command, and communication among peer staff will be used in the louvain algorithm. This information may look similar to the prison hierarchy chart posted at the beginning of the exercise.

• USE {Louvain algorithm}

Louvain algorithm is a method used to detect large communities in networks. It is heuristic, so it is able to do this very quickly. This also means it does not guarantee the best results. Even with this limitation, this solution paired with staff access to the dashboard and risk lists should provide ample communication. Louvain algorithm uses an iterative process to move nodes to adjacent communities (nodes) until modularity no longer increases. The louvain algorithm them uses this same process to connect **bigger** adjacent communities (super nodes) until modularity ceases to increase

- TO {maximize modularity and to detect large communities and facilitate weekly information about risk assessment and risk lists}

In a hierarchy, chain of command system laid out like the one for the IDOC, it is easy to envision how effective this would be in spreading important information like violent attack risk lists. Each staff node under a department head could be absorbed and combined into a super node including the department head. Select department head super nodes could then be absorbed into even bigger super nodes under assistant wardens, etc. A visual representation of what this might look like (so long as it resulted in increased modularity) is presented below.

Fig 1 Nodes combine to form super nodes

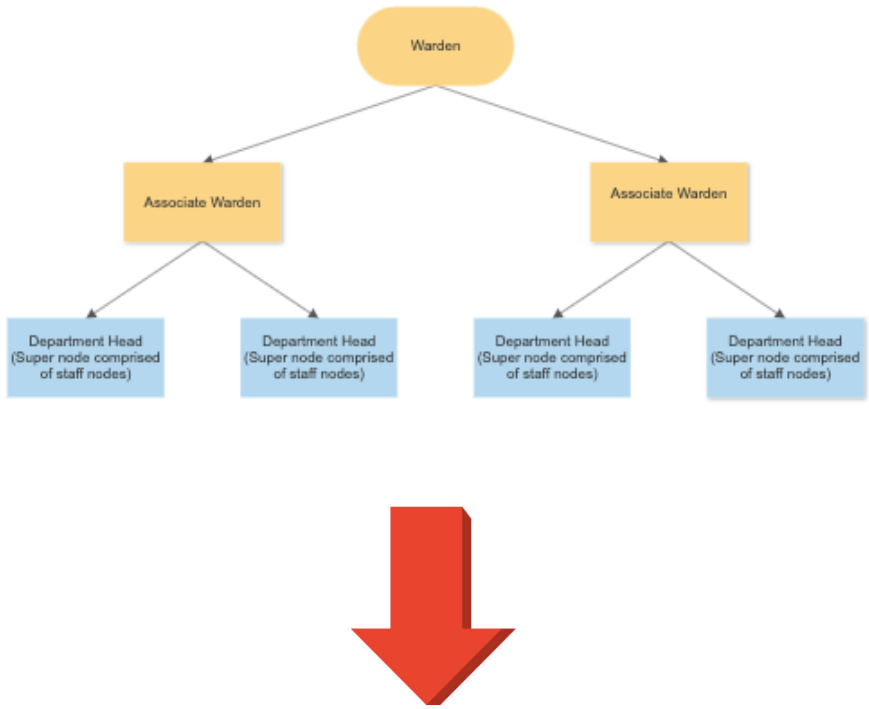
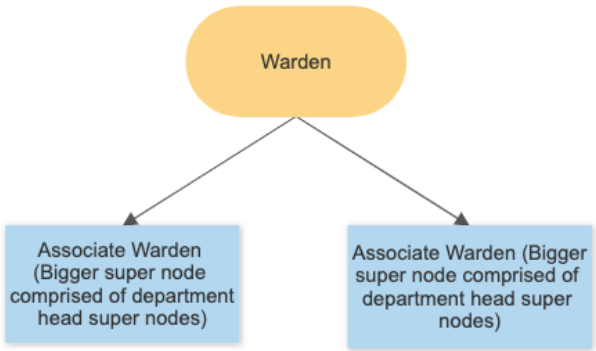


Fig 2 Super nodes combing to form bigger super nodes



Conclusion

It was interesting to read about the chosen solution of the IDOC. There are ways in which my proposed solution was similar (integrating of current data to form basis for new risk assessment tool and the use of weekly reports) and many more ways in which it was different (the use of conduct violations as outcome data, decision trees for models, some of the factors mentioned, etc).[4]

In the end, the new risk assessment tool created by the IDOC resulted in a 50% decrease in staff assaults in a 6 month testing period and a 20% decrease in inmate attacks on other inmates and created a number of programs and policies to run parallel with the new risk tool. In the world of analytics, where solutions are many, there is reason to be optimistic about one the IDOC implemented.

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[4] Information courtesy of https://www.sas.com/en_us/customers/idoc.html